Group B: Machine Learning

Laboratory Practice III – Oral / Viva Questions with Short Answers

SPPU – Fourth Year Computer Engineering (2019 Course)

Prepared from the uploaded syllabus: Group B Machine Learning practical assignments and the related Machine Learning course topics. The assignment list specifies any five assignments and one mini-project as mandatory.

1. Uber Fare Prediction – Preprocessing, Linear Regression and Random Forest

Q1. What is regression?

Answer: Regression is a supervised learning task used to predict a continuous numerical value, such as a fare or price.

Q2. What is the target variable in Uber fare prediction?

Answer: The target is the fare amount to be predicted.

Q3. What is a feature?

Answer: A feature is an input variable used by a model to make a prediction, such as pickup coordinates, drop-off coordinates, or passenger count.

Q4. What is data preprocessing?

Answer: It is the process of cleaning and transforming raw data before using it to train a machine learning model.

Q5. Why is preprocessing required?

Answer: It improves data quality and helps models learn from consistent, usable data.

Q6. What are missing values?

Answer: Missing values are entries where a feature value is not available or recorded.

Q7. How can missing values be handled?

Answer: They can be removed or filled using values such as the mean, median, or mode, depending on the data.

Q8. What is an outlier?

Answer: An outlier is a data point that is unusually far from most other observations.

Q9. Why should outliers be identified?

Answer: They can distort statistical measures and affect model training and evaluation.

Q10. What is correlation?

Answer: Correlation measures the strength and direction of association between two variables.

Q11. What is Linear Regression?

Answer: A supervised learning algorithm that models the relationship between input features and a continuous target using a linear equation.

Q12. What is the equation of simple Linear Regression?

Answer: $y = mx + c$, where m is the slope and c is the intercept.

Q13. What is Random Forest Regression?

Answer: It is an ensemble method that combines predictions from multiple decision trees, commonly by averaging them.

Q14. What is R-squared (R^2)?

Answer: It measures how much of the variation in the target is explained by the model; a higher value is generally better.

Q15. What is RMSE?

Answer: Root Mean Squared Error is the square root of the average squared prediction error; lower values indicate smaller errors.

2. Email Spam Detection – KNN and SVM

Q1. What is classification?

Answer: Classification is supervised learning used to assign an input to a class or category.

Q2. What are the two classes in spam detection?

Answer: Spam and not spam (normal or ham).

Q3. What is binary classification?

Answer: A classification problem with exactly two possible classes.

Q4. What is K-Nearest Neighbors (KNN)?

Answer: KNN predicts a class using the majority class among the K closest training examples.

Q5. What does K represent in KNN?

Answer: K is the number of nearest neighbours considered when making a prediction.

Q6. What happens if K is too small?

Answer: The model can become sensitive to noise and may overfit.

Q7. What happens if K is too large?

Answer: The model may become too general and can underfit.

Q8. Why is feature scaling important for KNN?

Answer: KNN uses distances, so features with larger numeric ranges can dominate unless features are scaled.

Q9. What is SVM?

Answer: Support Vector Machine is a supervised learning algorithm that finds a decision boundary separating classes.

Q10. What is a hyperplane?

Answer: A decision boundary that separates classes in the feature space.

Q11. What are support vectors?

Answer: The training points closest to the decision boundary; they help determine the separating boundary.

Q12. What is a kernel in SVM?

Answer: A function that allows SVM to model non-linear decision boundaries by representing data in another feature space.

Q13. What is accuracy?

Answer: The fraction of all predictions that are correct.

Q14. What is a confusion matrix?

Answer: A table comparing actual and predicted classes, showing true positives, true negatives, false positives and false negatives.

Q15. Why can accuracy be misleading for spam detection?

Answer: If classes are imbalanced, a model can achieve high accuracy while performing poorly on the minority class.

3. Bank Customer Churn – Neural Network Classifier

Q1. What is customer churn?

Answer: Customer churn means a customer stops using or leaves a company's service.

Q2. What is a neural network?

Answer: A machine learning model made of connected processing units called neurons, arranged in layers.

Q3. What are the main layers in a basic neural network?

Answer: Input layer, one or more hidden layers, and output layer.

Q4. What is an input layer?

Answer: The layer that receives the feature values.

Q5. What is a hidden layer?

Answer: A layer that transforms inputs using weights, biases and activation functions.

Q6. What is an output layer?

Answer: The layer that produces the model's prediction.

Q7. What is an activation function?

Answer: A function applied to a neuron's output to introduce non-linearity.

Q8. Name common activation functions.

Answer: ReLU, sigmoid and tanh.

Q9. What is the sigmoid function commonly used for?

Answer: It maps a value to a range between 0 and 1 and is often used for binary classification output.

Q10. Why split data into training and testing sets?

Answer: Training data is used to fit the model; test data evaluates how well it performs on unseen examples.

Q11. What is normalization?

Answer: Scaling feature values to a common range or scale.

Q12. Why normalize data for a neural network?

Answer: It can improve training stability and help optimisation converge more effectively.

Q13. What is an epoch?

Answer: One complete pass through the training dataset.

Q14. What is a loss function?

Answer: A function that measures the difference between model predictions and actual values.

Q15. What is a confusion matrix used for?

Answer: It shows correct and incorrect predictions for each class.

4. Gradient Descent – Finding a Local Minimum

Q1. What is Gradient Descent?

Answer: An iterative optimisation algorithm that updates parameters in the direction that reduces a function's value.

Q2. What is the gradient?

Answer: The derivative, or vector of partial derivatives, that indicates the direction of steepest increase.

Q3. In which direction does Gradient Descent move?

Answer: In the negative gradient direction.

Q4. What is the learning rate?

Answer: A parameter that controls the size of each update step.

Q5. What happens if the learning rate is too high?

Answer: Updates may overshoot the minimum or fail to converge.

Q6. What happens if the learning rate is too low?

Answer: Convergence can be very slow.

Q7. What is the update rule for one variable?

Answer: x_new = x_old - learning_rate × derivative at x_old.

Q8. For $y = (x + 3)^2$, what is the derivative?

Answer: $dy/dx = 2(x + 3)$.

Q9. Where is the minimum of $y = (x + 3)^2$?

Answer: At x = -3, where the function value is 0.

Q10. What is the difference between a local and global minimum?

Answer: A local minimum is lowest in a nearby region; a global minimum is lowest over the entire domain.

5. KNN on Diabetes Dataset – Evaluation Metrics

Q1. What is the purpose of the diabetes dataset practical?

Answer: To use KNN to classify the target outcome and evaluate the classifier.

Q2. What is supervised learning?

Answer: Learning from labelled examples, where each training example includes its correct target.

Q3. What is a training dataset?

Answer: The data used to fit or train a machine learning model.

Q4. What is a test dataset?

Answer: Held-out data used to evaluate the trained model.

Q5. What is a true positive (TP)?

Answer: A positive case correctly predicted as positive.

Q6. What is a true negative (TN)?

Answer: A negative case correctly predicted as negative.

Q7. What is a false positive (FP)?

Answer: A negative case incorrectly predicted as positive.

Q8. What is a false negative (FN)?

Answer: A positive case incorrectly predicted as negative.

Q9. How is accuracy calculated?

Answer: (TP + TN) / (TP + TN + FP + FN).

Q10. How is precision calculated?

Answer: TP / (TP + FP). It measures how many predicted positives are actually positive.

Q11. How is recall calculated?

Answer: TP / (TP + FN). It measures how many actual positives are correctly identified.

Q12. What is the error rate?

Answer: The fraction of predictions that are incorrect; it is 1 - accuracy.

Q13. What is the F1-score?

Answer: The harmonic mean of precision and recall: 2 × precision × recall / (precision + recall).

Q14. Why use a confusion matrix?

Answer: It provides the counts needed to understand different types of classification errors.

6. K-Means / Hierarchical Clustering – Sales Dataset

Q1. What is clustering?

Answer: An unsupervised learning technique that groups similar data points together.

Q2. What is the difference between supervised and unsupervised learning?

Answer: Supervised learning uses labelled data; unsupervised learning finds patterns in data without target labels.

Q3. What is K-Means clustering?

Answer: An algorithm that partitions data into K clusters by assigning points to their nearest centroid and updating the centroids.

Q4. What does K represent in K-Means?

Answer: The number of clusters to form.

Q5. What is a centroid?

Answer: The mean position of the data points assigned to a cluster.

Q6. What are the main steps of K-Means?

Answer: Choose K centroids, assign each point to the nearest centroid, recompute centroids, and repeat until assignments stabilise.

Q7. Why is feature scaling important in clustering?

Answer: Distance-based clustering can be dominated by features with larger numeric ranges.

Q8. What is the elbow method?

Answer: A method for choosing K by plotting within-cluster sum of squares against K and looking for a bend or elbow.

Q9. What is WCSS?

Answer: Within-Cluster Sum of Squares; it measures the squared distances of points from their assigned cluster centroid.

Q10. What is hierarchical clustering?

Answer: A clustering method that builds a hierarchy of clusters, often represented using a dendrogram.

Q11. What is a dendrogram?

Answer: A tree diagram showing how individual points or clusters are merged or split.

Q12. What is the difference between agglomerative and divisive clustering?

Answer: Agglomerative clustering starts with individual points and merges them; divisive clustering starts with one cluster and splits it.

Q13. Does K-Means require the number of clusters in advance?

Answer: Yes, K must be specified before running the basic K-Means algorithm.

Q14. Is clustering supervised or unsupervised?

Answer: Unsupervised.

7. General Machine Learning Concepts from the Theory Syllabus

Q1. What is Machine Learning?

Answer: A field of AI in which systems learn patterns from data to make predictions or decisions.

Q2. What is the difference between AI, ML and Data Science?

Answer: AI is the broader goal of intelligent systems; ML is a set of methods that learn from data; Data Science uses data analysis and related methods to extract insights.

Q3. Name the main types of machine learning listed in the syllabus.

Answer: Supervised, unsupervised, semi-supervised and reinforcement learning.

Q4. What is overfitting?

Answer: When a model learns training data too closely, including noise, and performs poorly on unseen data.

Q5. What is underfitting?

Answer: When a model is too simple to capture the patterns in the data.

Q6. What is bias?

Answer: Error caused by overly simple assumptions in a model.

Q7. What is variance?

Answer: Sensitivity of a model's predictions to changes in the training data.

Q8. What is feature engineering?

Answer: Creating, transforming or selecting useful input features for a machine learning model.

Q9. What is PCA?

Answer: Principal Component Analysis is a dimensionality reduction technique that transforms data into principal components capturing as much variance as possible.

Q10. What is cross-validation?

Answer: A method of evaluating a model by training and validating it on different splits of the dataset.

Q11. What is ensemble learning?

Answer: Combining multiple models to improve prediction performance or robustness.

Q12. What is Random Forest?

Answer: An ensemble of decision trees whose predictions are combined; for regression, predictions are commonly averaged.

Q13. What is dimensionality reduction?

Answer: Reducing the number of input features while trying to preserve useful information.

Q14. What is reinforcement learning?

Answer: A learning approach in which an agent learns by interacting with an environment and receiving rewards or penalties.

Quick Revision: Evaluation Metrics

Accuracy = (TP + TN) / (TP + TN + FP + FN)
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1-score = 2 × (Precision × Recall) / (Precision + Recall)
Error rate = 1 – Accuracy
RMSE = square root of the mean squared prediction error.
R^2 = coefficient of determination.

Practical exam tip: Be prepared to explain the dataset, input features, target variable, preprocessing steps, train/test split, algorithm used, output metrics and what the result means. For numerical examples, follow the exact dataset and code given by your instructor.

Scope note: The PDF focuses on the six Group B practical assignments listed in the uploaded syllabus, with a final section of supporting theory topics. The syllabus also lists mini-projects on stock prediction, Titanic survival prediction and signature identification.